

Project 2 - Safe Combination Lock V1.0:

The objective of this circuit is to only work “correctly” when the right combination of gates is selected (correct password), displaying a green light that confirms the selection; otherwise, the circuit will display a red light, indicating the “password” is incorrect. Additionally, in order to display the result of the combination selected, the user has to push a button to reveal the result, and light up the LED.

The first version of the design was simply a single LED red light that indicated that the password was correct, but then I decided to integrate a green light for confirmation and leave the red light for error indication. Later I integrated the push button in order to save battery and a better user experience.

The following truth table depicts the functionality of the circuit:
(for the LEDs 1 = ON; 0 = OFF)

Row	A B C D	Red LED	Green LED
0	0000	1	0
1	0001	1	0
2	0010	1	0
3	0011	1	0
4	0100	1	0
5	0101	1	0
6	0110	1	0
7	0111	1	0
8	1000	1	0
9	1001	1	0
10	1010	1	0
11	1011	0	1
12	1100	1	0
13	1101	1	0
14	1110	1	0
15	1111	1	0

Red LED K-Map: Maxterm Expression

$$F(A,B,C,D) = \prod M(11) = (A' + B + C' + D')$$

AB CD					
		00	01	11	10
00	0	1	1	1	1
01	1	1	1	1	1
11	3	1	1	1	0
10	2	1	1	1	1

Green LED K-map: Minterm Expression

$$F(A,B,C,D) = \sum m(11) = AB'CD$$

AB CD					
		00	01	11	10
00	0	0	0	0	0
01	1	0	0	0	0
11	3	0	0	0	1
10	2	0	0	0	0

In order to build this circuit, I used the following elements:

- 7408 Quad AND gate (1)
- 7404 Hex Inverter (1)
- DIP Switch (1)
- 220-ohm resistors (6)
- Green LED (1)
- Red LED (1)
- Pushbutton (1)
- Breadboard (1)

Further Improvements:

- The lock could stay in a constant Stand By mode, where the display is not showing any lights.
- Numeral Pad: instead of using a DIP Switch, I could implement a numeral pad so the user can select the numbers instead of moving switches.
- 7 Segment Displays: a display that shows the numbers selected in decimal
- Timer: a new segment for this circuit could be a timer that starts running when the user picks the first 4 digits, and